



Here's the latest CodeChef.com Challenge, your monthly dose of coding puzzles from India's biggest coding contest!

Solve this month's puzzle, and you could be one of the three lucky people to win a cash prize of Rs 1,000 each!

The August edition's puzzle:

There is a village situated on a hill. It doesn't have any resources of its own and hence everything has to be brought up from the city market. Every month the government sends a cart down to the market to buy what is needed by the people. While returning to the village, the cart gets overloaded every time. The cart is pulled by one horse at a time. One horse pulls the cart for 10 km and can be used only once in a day. There were three horses initially and after using them, the cart has to stop so that the horses get ready for the next day. But due to the load and inefficient braking system, the cart slides back 10 km. During the last day of the journey, one more horse joins the trio to help pull the cart. How many days will it take the cart to reach the village if the distance between the market and the village is 400 km?

The Solution:

According to the problem, three horses are used initially, so the total distance covered by them will be
 $= 3 \times 10 = 30$ km
 But due to the slope and inefficient braking system, the cart slides back 10 km. So, for each day with the 3 horses, the cart can travel $30 - 10$ km = 20 km ...
 Eqn (1)
 Considering the last day's scenario, there were 4 horses so the cart got pulled 4×10 km that day. So the distance travelled on the last day = 40 km ...
 Eqn (2)
 Distance left = total distance – distance travelled on the last day

$$= 400 - 40$$

$$= 360 \text{ km}$$

The days required to cover the initial 360 km = $360/20 = 18$ days (using Eqn 1)

And on the last day, the distance covered = 40 km (using Eqn 2)
 Hence, the total number of days taken = 19 .

And the winners this month are:

- Satyabrata Satpathy
- Syed Mohammed Khawar
- Ashish Monhe

Here's your CodeChef Puzzle of the Month:

There is a 50 cm long line of ants moving ahead. The last ant in the line had an urgent message to give to the queen ant, which is moving at the first position. So while the line is moving, it runs ahead, reaches the first ant and conveys the message and without stopping, it runs and comes back to its original position. (Consider that no time is lost in giving the message.) Meanwhile the whole line has moved ahead by 50 cm.

The question is how much distance did the ant with the message cover in that time—assuming that it ran the whole distance at a uniform speed.

Looking for a little extra cash?

Solve this month's CodeChef Challenge puzzle and send in your answer to codechef@efyindia.com or ify@codechef.com by September 15, 2012. Three lucky winners get to win some awesome prizes both in cash and other merchandise! You can also participate in the contest by visiting the online space for the article at <https://www.facebook.com/LinuxForYou> 

About CodeChef: CodeChef.com is India's first, non-commercial, online programming competition, featuring monthly contests in more than 35 different programming languages. Log on to CodeChef.com for the September Challenge that takes place from the 1st to the 11th, to win cash prizes up to Rs 20,000. So keep visiting CodeChef to participate in the multiple programming contests that take place on the website throughout the month!